Abstract

This experiment studies the design and implementation of adder and comparator circuits using logic gates. Hardware implementation and simulation were performed to verify theoretical results.

Objective

To design, implement, simulate, and verify adder and comparator circuits using basic logic gates and compare theoretical and practical outputs.

Introduction

Adder and comparator circuits are fundamental combinational logic circuits used in digital systems. Adders perform binary addition and are key components of arithmetic logic units (ALUs), while comparators determine whether one binary number is equal to, greater than, or less than another. In this experiment, half adder, full adder, and magnitude comparator circuits were designed, implemented on a digital trainer board, and verified through simulation and hardware testing to strengthen understanding of Boolean logic and digital circuit design.

Theory: An adder is a combinational logic circuit that performs addition of binary numbers. Adders are fundamental building blocks of digital systems and are essential in the Arithmetic Logic Unit (ALU) of processors. There are mainly two kinds of adders, half adder and full adder.

1. Half Adder

A half adder adds two single-bit binary inputs.

Inputs: A, B

Outputs: Sum (S), Carry (Cout)

Boolean expressions:

- $S = A \oplus B$
- $Cout = AB$

2. Full Adder: A full adder adds three inputs- Two binary bits and one carry input from a previous stage

Inputs:
A, B, Cin

Outputs:
Sum (S), Carry (Cout)

Boolean expressions:

- $S = A \oplus B \oplus Cin$
- $Cout = Cin(A \oplus B) + AB$

3. Multi-bit Binary Addition: Multiple full adders can be connected in cascade to add multi-bit numbers. Example:

- LSB stage receives Cin = 0
- Carry output of one stage becomes carry input of the next stage
- This structure forms an n-bit adder

Magnitude Comparator: A magnitude comparator is a combinational circuit that compares two binary numbers and determines their relationship. Outputs of comparator for two inputs A and B-

- $A = B \rightarrow$ numbers are equal
- $A > B \rightarrow$ A is greater
- $A < B \rightarrow$ A is smaller

1. 1-Bit Comparator Logic

Conditions:

- A = B when both inputs are same (00 or 11)
- A > B when A=1, B=0
- A < B when A=0, B=1

Boolean expressions:

- $(A = B) = A'B' + AB$
- $(A < B) = A'B$
- $(A > B) = AB'$

2. Multi-bit Comparator: Multi-bit comparators are formed by combining multiple 1-bit comparator blocks. They compare binary words bit by bit starting from the most significant bit.

Apparatus:

1. Digital trainer board
2. IC 7408:2 pcs
3. IC 7404:2 pcs
4. IC 7486:2 pcs
5. IC 7431:2 pcs
6. IC 7483:1 pcs
7. Connecting wires

Experimental Procedure:

1. First, determine the expected outputs and complete the truth table for the full adder circuit from the theory section. This helps you know what results you should observe during the experiment.

2. Next, identify which logic gates are required to build the circuit and note the corresponding IC numbers. Check that all components are available and working properly before starting the setup.

3. Place the ICs carefully on the digital trainer board. Connect the power supply by applying +5V and ground to the appropriate pins of each IC.

4. After powering the circuit, connect the components using wires according to the logic diagram provided in the lab sheet. Attach the output terminals to LEDs so the results can be observed visually.

5. Apply different combinations of input values using the toggle switches. Observe the LED outputs for each input combination and record the results in a table.

6. Finally, compare the observed experimental outputs with the theoretical truth table and simulation results. Note any differences and analyze possible reasons for the variation.

Data Table

Truth Table (Full Adder)

A	B	Cin	S	C

0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Circuit Implementation

Full Adder

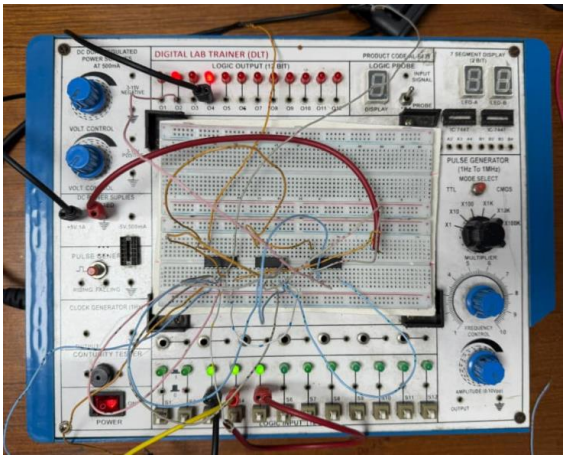


Fig-1: When A=B=C=1

Comparator

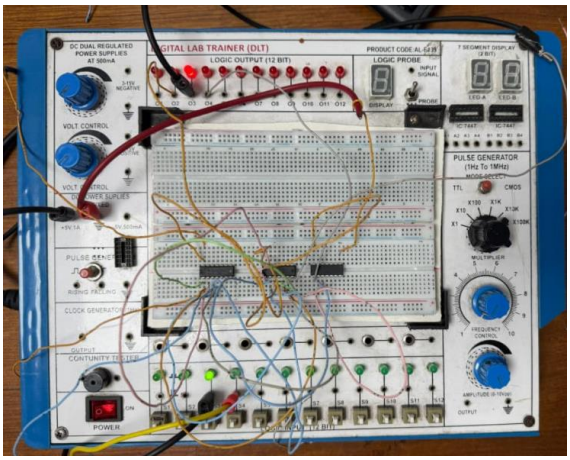


Fig-4: When A=1, B=0

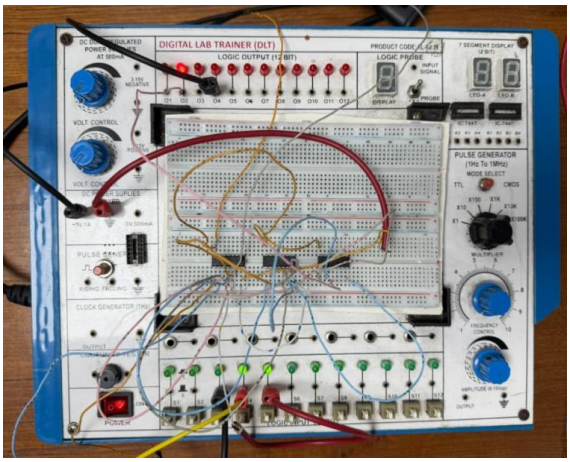


Fig-2: When A=0, B=C=1

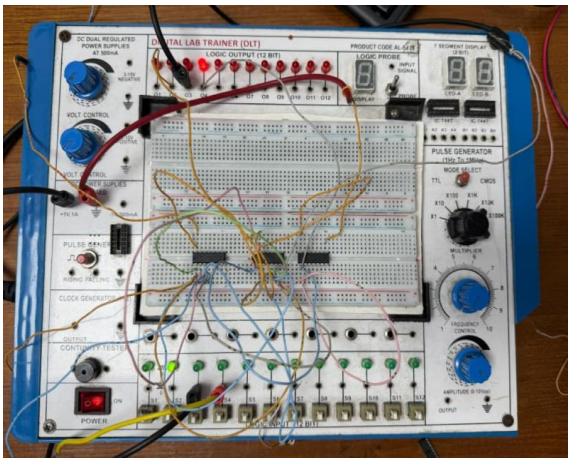


Fig-5: When A=B=0

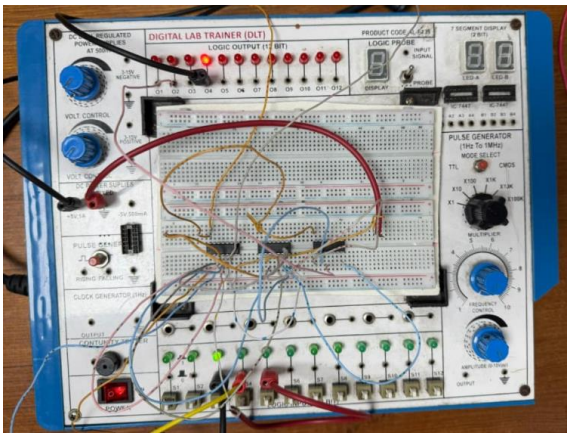
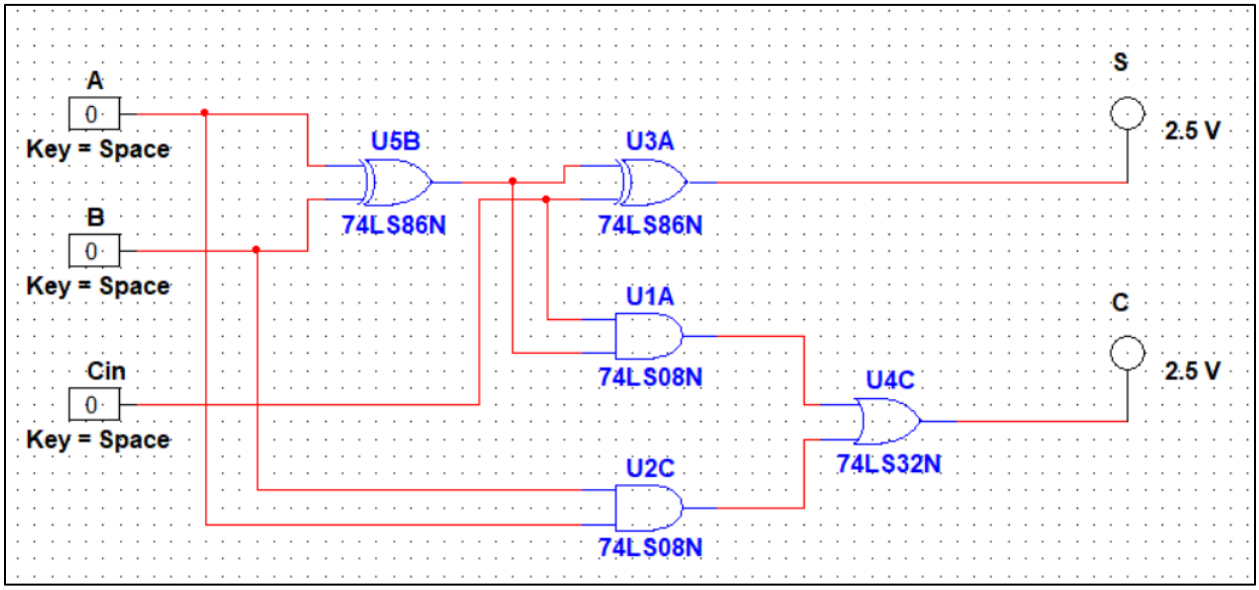


Fig-3: When A=1, B=C=0

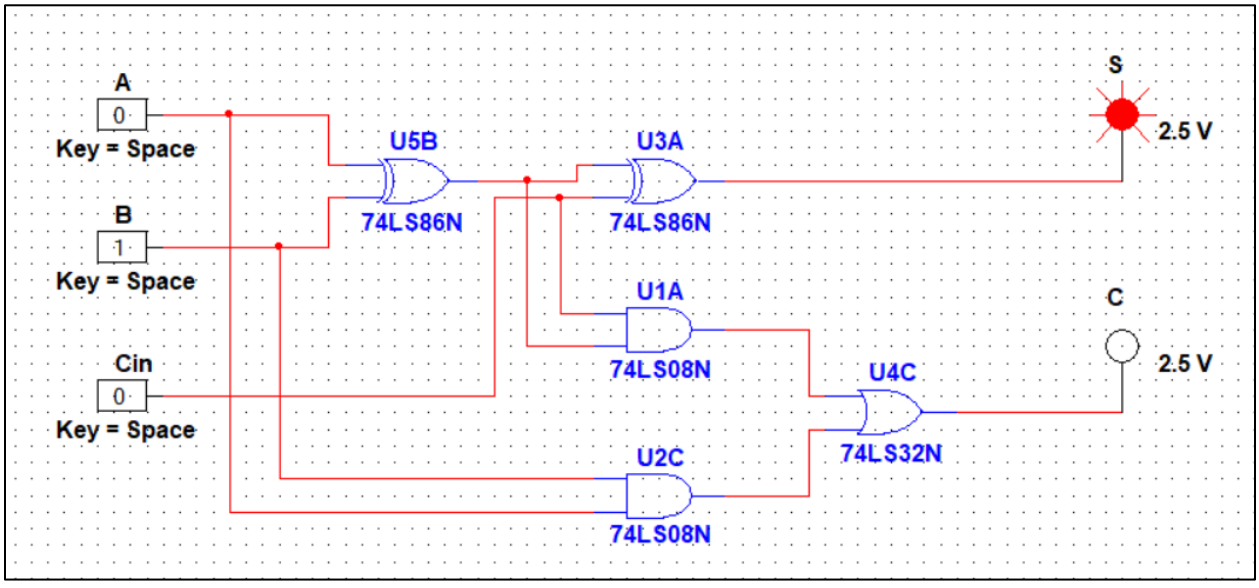
Simulation and Measurement

Full Adder

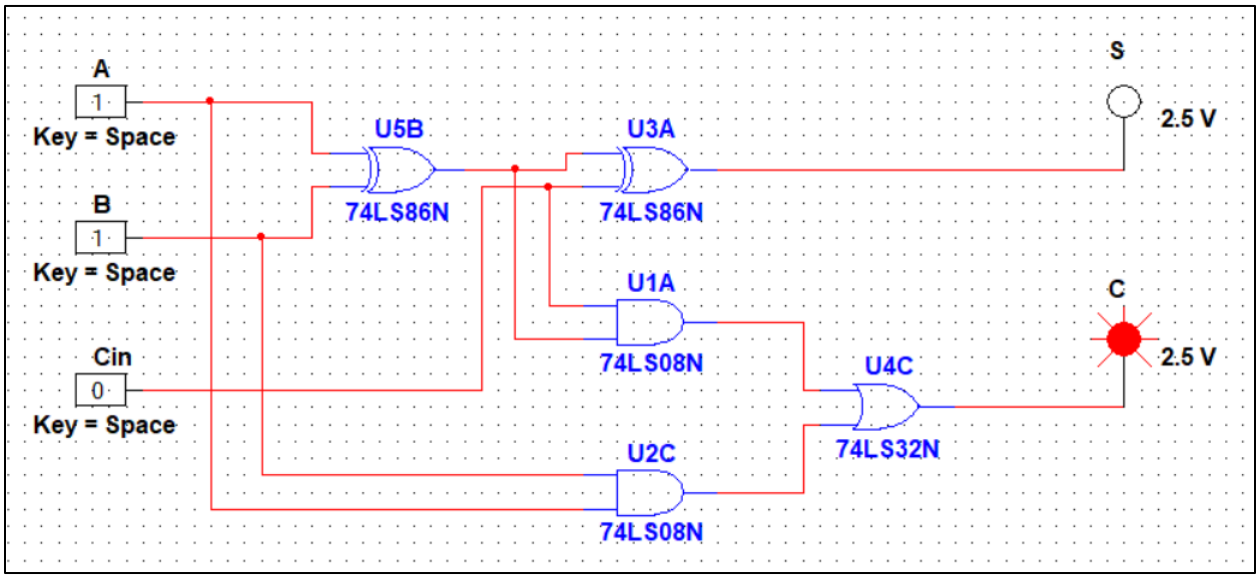
When A=B=Cin=0.



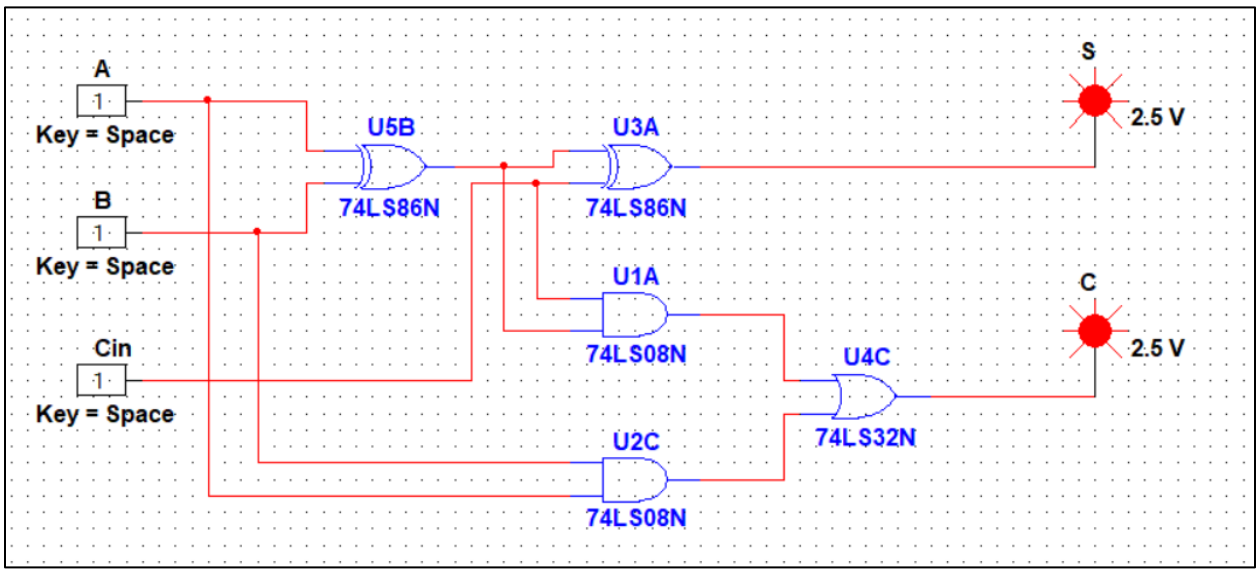
When A=0, B=1, Cin=0.



When A=B=1, Cin=0.

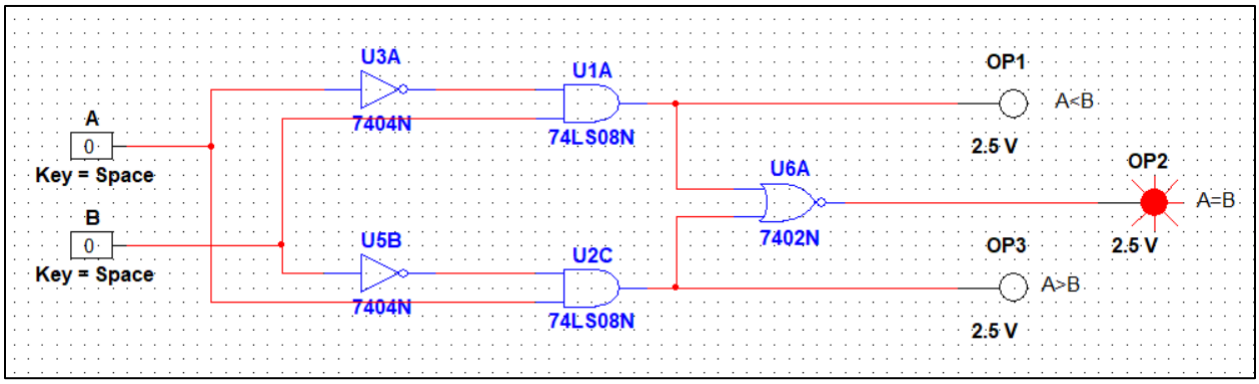


When A=B=Cin=1.

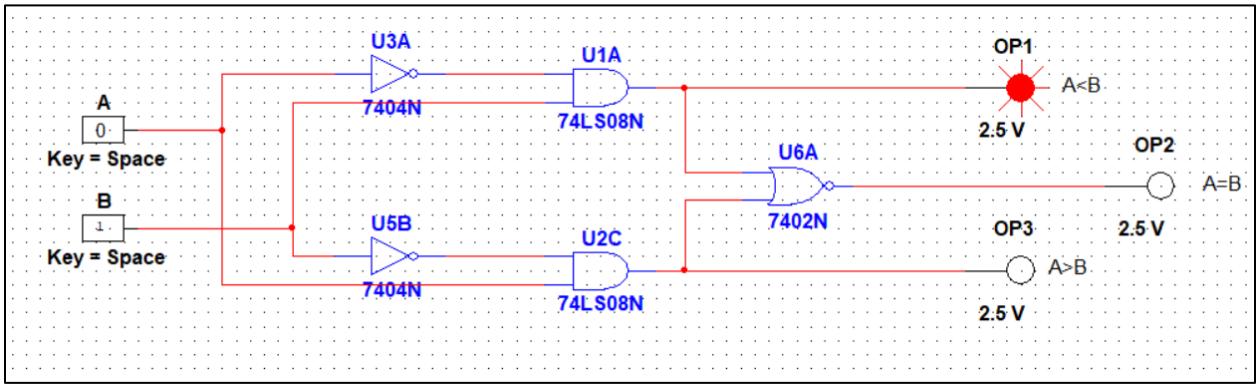


Comparator

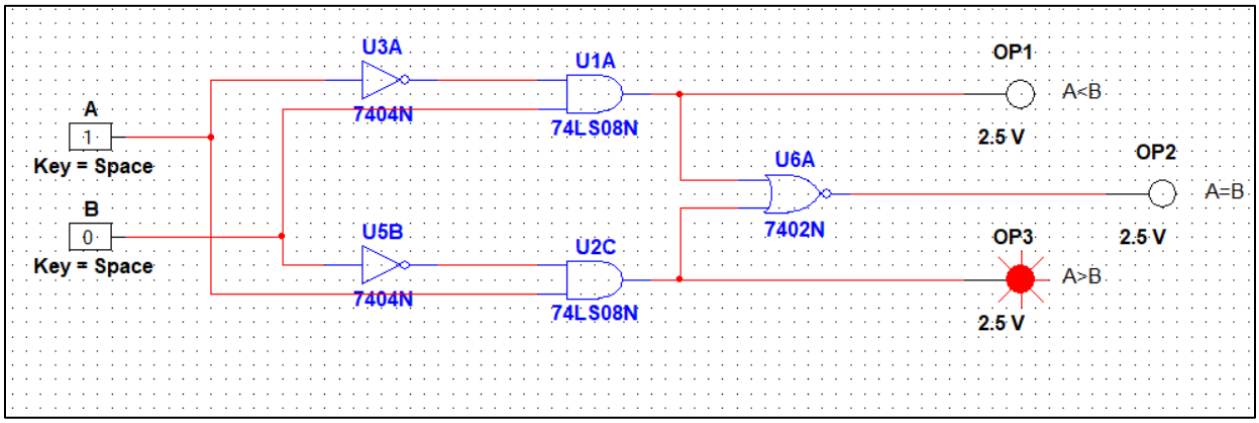
When A=B=0.



When A=0, B=1.



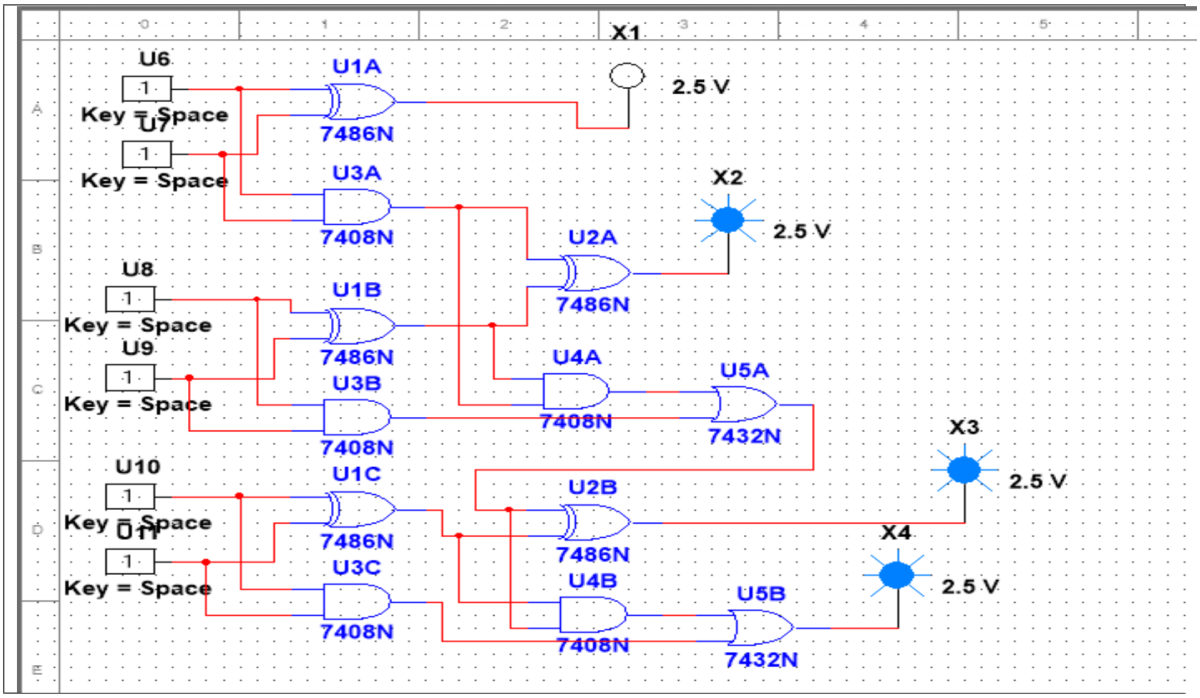
When A=1, B=0



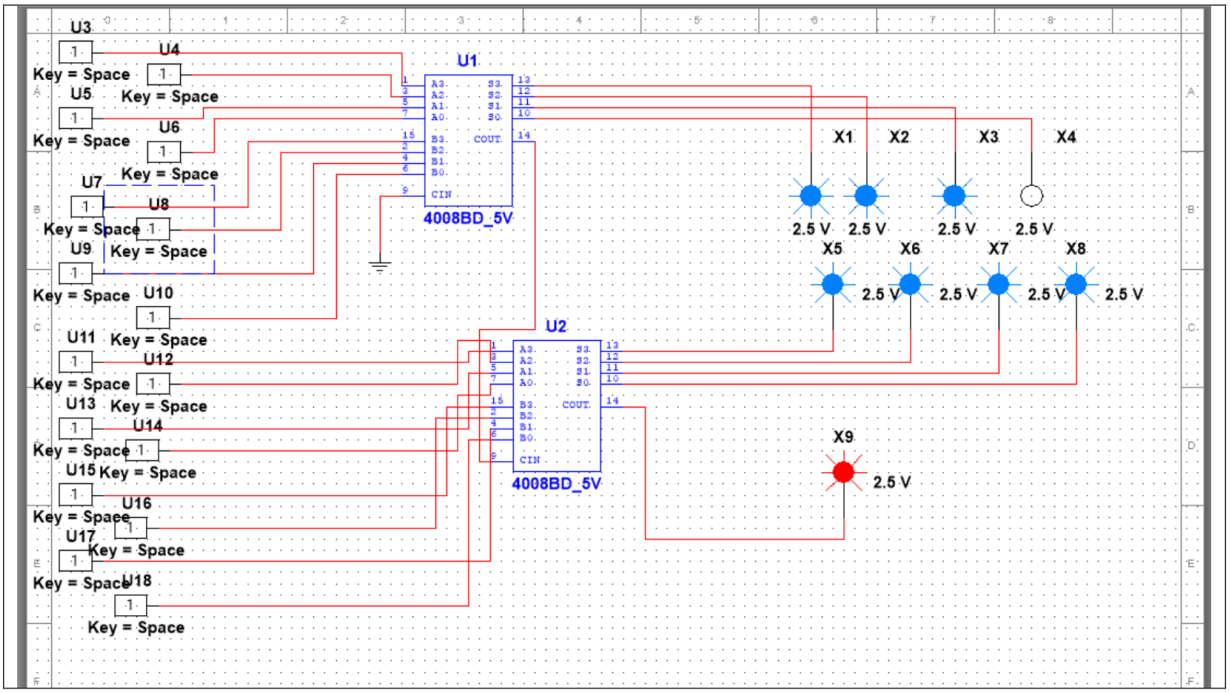
Report Questions:

Question 1: Design a full adder circuit for performing 3-bit binary addition from Multisim.

adder



Question 2: Design an 8-bit full adder using a 4-bit full adder IC 4008 from Multisim.



Discussion: In this experiment, adder and comparator circuits were designed and implemented using basic logic gates. The half adder and full adder circuits were tested for all possible input combinations, and the observed outputs closely matched the theoretical truth tables. This confirms the correctness of the Boolean expressions used in the design. Multi-bit addition was also verified

by cascading full adders, where the carry output of one stage was successfully used as the carry input of the next stage.

For the comparator circuit, the 1-bit magnitude comparator was implemented using NOT, AND, and OR gates based on the SOP expressions. During simulation in NI Multisim, a wiring and gate-selection error was initially observed, which caused incorrect voltage levels at the output. After correcting the mistake by using the proper gate configuration, the comparator produced correct outputs for the conditions A=B, A>B, and A<B. Minor deviations such as floating voltage levels were mainly due to incorrect connections and output shorting, which were resolved through careful debugging.

Conclusion: The experiment was successfully completed and met its intended objectives. The design and operation of adder and comparator circuits were clearly understood through theoretical analysis, hardware implementation, and simulation. The results obtained from the practical setup and Multisim simulation were consistent with the expected theoretical values. This experiment enhanced understanding of combinational logic circuits and demonstrated the importance of correct gate selection and proper wiring in digital circuit design.